

Multi-Objective Bayesian Optimization and Independent Pareto Candidate Verification for a 200 MeV Electron Injector Linac

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July 24, 2026

Abstract

High-brightness electron injector linacs require simultaneous minimization of transverse emittances ($\varepsilon_{n,x}, \varepsilon_{n,y}$) and RMS energy spread (σ_E) under stringent beam quality and transmission constraints. Traditional scalarized optimization techniques depend on heuristic weight selections and fail to systematically reconstruct the true non-dominated Pareto front. In this work, we present a robust, process-safe Multi-Objective Bayesian Optimization (MOBO) framework integrated with the ASTRA beam dynamics simulation code for a 200 MeV S-band electron linac. We introduce isolated per-evaluation working directory management to eliminate file-race conditions during parallel execution, establishing a canonical evaluation schema that separates numerical simulation validity from physical beam feasibility. Using quasi-Monte Carlo Sobol initialization and q -logarithmic Noisy Expected Hypervolume Improvement (q LogNEHVI) acquisition functions with a fixed reporting reference point standard, we compare unconstrained and feasibility-constrained MOBO performance. Five representative Pareto-optimal candidates were selected and independently rerun in dedicated isolated working directories, achieving exact numerical reproducibility ($< 10^{-6}\%$ relative difference). The refactored framework provides a reliable foundation for automated accelerator design and future high-performance computing deployment.

Keywords — Electron Injector Linac, Multi-Objective Bayesian Optimization, ASTRA Beam Dynamics, Pareto Front Verification, q LogNEHVI, Hypervolume.

1 Introduction

Modern linear accelerators (linacs), such as those serving X-ray Free-Electron Lasers (XFELs) and ultrafast electron diffraction (UED) facilities, demand high-peak-current, low-emittance electron beams with minimal energy spread [1, 2]. Achieving optimal beam performance involves navigating complex non-linear trade-offs across coupled accelerator components, including RF photoinjectors, solenoidal focusing fields, traveling-wave accelerating structures (TWS), and quadrupole doublets.

Historically, scalarized optimization techniques (e.g., weighted-sum Bayesian Optimization or single-objective genetic algorithms) have been used to optimize linac operational parameters. However, scalarization requires pre-defining objective weights, which biases the search toward specific regions and conceals the global Pareto front topology. Multi-Objective Bayesian Optimization (MOBO) using Gaussian Process (GP) surrogates offers an efficient alternative by directly learning the non-dominated Pareto front while minimizing expensive particle tracking simulations [3, 4].

In this paper, we present the production-grade refactoring and experimental validation of a MOBO framework applied to a 200 MeV S-band electron injector linac simulated via ASTRA [2]. The main contributions of this study are:

1. **Process-Safe Directory Isolation:** Operating every ASTRA simulation inside a dedicated evaluation directory (`results/<run_id>/work/eval_<id>/`), eliminating file collisions in multi-core parallel environments.
2. **Strict Failure Semantics:** Explicitly separating numerical simulation convergence from physical beam feasibility, preventing surrogate model distortion from artificial sentinel values.
3. **Fixed Hypervolume Reporting:** Standardizing fixed reporting reference points across optimization iterations to enable monotonic, un-biased hypervolume growth tracking.
4. **Independent Pareto Verification:** Independent re-simulation of key Pareto candidates to guarantee absolute data integrity and zero cross-talk.

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2 Linac System and Problem Formulation

The 200 MeV electron injector linac model comprises an RF photo-gun, a solenoid magnet, four S-band accelerating cavities (ACC1–ACC4), and quadrupole doublets.

2.1 Design Variables

The optimization space consists of 6 continuous design parameters ($D = 6$):

$$\mathbf{x} = [B_{\text{sol}}, G_{q1}, G_{q2}, \phi_{\text{gun}}, \phi_{\text{acc1/2}}, \phi_{\text{acc3/4}}]^\top \in \mathcal{X} \subset \mathbb{R}^6 \quad (1)$$

where B_{sol} is the solenoid peak magnetic field, G_{q1} and G_{q2} are quadrupole gradients, ϕ_{gun} is the RF gun phase, $\phi_{\text{acc1/2}}$ controls the coupled phase of ACC1 and ACC2, and $\phi_{\text{acc3/4}}$ controls the coupled phase of ACC3 and ACC4. The parameter search bounds are summarized in Table 1.

Table 1: Design variable bounds and nominal values for the 200 MeV injector linac.

Variable	ASTRA Key	Nominal	Lower Bound	Upper Bound	Unit
B_{sol}	<code>solenoid:maxb(1)</code>	0.1948	0.0974	0.2922	T
G_{q1}	<code>quadrupole:q_grad(1)</code>	1.2865	0.6433	1.9298	T/m
G_{q2}	<code>quadrupole:q_grad(2)</code>	-2.8867	-4.3300	-1.4433	T/m
ϕ_{gun}	<code>cavity:phi(1)</code>	35.62	32.06	39.19	deg
$\phi_{\text{acc1/2}}$	<code>cavity:phi(2,3)</code>	-39.51	-43.46	-35.56	deg
$\phi_{\text{acc3/4}}$	<code>cavity:phi(4,5)</code>	310.05	279.05	341.06	deg

2.2 Objective Functions

We simultaneously minimize three beam quality metrics measured at the end of the linac ($s \approx 16.2$ m):

$$\mathbf{y}_{\text{phys}}(\mathbf{x}) = [\varepsilon_{n,x}(\mathbf{x}), \varepsilon_{n,y}(\mathbf{x}), \sigma_E(\mathbf{x})]^\top \quad (2)$$

where $\varepsilon_{n,x}$ and $\varepsilon_{n,y}$ are normalized RMS horizontal and vertical emittances ($\text{m} \cdot \text{rad}$), and σ_E is the RMS energy spread (eV). To interface with BoTorch [5], physical objectives are converted to model space via canonical negation: $\mathbf{y}_{\text{model}} = -\mathbf{y}_{\text{phys}}$.

2.3 Beam Quality Constraints

Beams must satisfy 8 physical constraints to be classified as physically feasible:

$$\sigma_x, \sigma_y \leq 1.0 \text{ mm}, \quad \sigma_{x'}, \sigma_{y'} \leq 1.0 \text{ mrad}, \quad (3)$$

$$\sigma_z \leq 1.0 \text{ mm}, \quad E_{\text{kin}} \in [195.0, 205.0] \text{ MeV}, \quad \eta_{\text{trans}} \geq 90\% \quad (4)$$

3 MOBO Architecture and Reliability Framework

3.1 Directory Isolation and Parallel Evaluation

Running parallel ASTRA instances in a single directory leads to file overwrites (e.g., `astra.Log`, `astra.out`). We resolve this by instantiating an isolated work directory for each evaluation i :

$$\mathcal{D}_i = \text{results}/\langle \text{run_id} \rangle/\text{work}/\text{eval_00000i}/ \quad (5)$$

Every evaluation directory contains clean copies of input templates (`gun.dat`, `PAL_SOL_A.dat`, `TWS_Sband.dat`, `pal_photo2.ini`) and an independently generated `astra.in`. Parallel evaluations execute via `ProcessPoolExecutor` with process isolation.

3.2 Surrogate Modeling and Acquisition Optimization

Independent Gaussian Process (GP) surrogates with Matérn-5/2 kernels model each objective. Candidate batches of size $q = 4$ are selected by optimizing the $q\text{LogNEHVI}$ acquisition function:

$$\alpha_{q\text{LogNEHVI}}(\mathbf{X}) = \mathbb{E} [\log \text{HV} (\mathbf{Y}(\mathbf{X}) \cup \mathcal{P} \mid \mathbf{r}_{\text{acq}})] \quad (6)$$

In constrained MOBO (Phase 3), the acquisition function is weighted by the joint Probability of Feasibility $P(\text{feasible} \mid \mathbf{x})$.

4 Experimental Results and Discussion

Controlled campaigns were executed comparing Phase 2 (Unconstrained MOBO) and Phase 3 (Constrained MOBO) under identical random seeds ($N_{\text{init}} = 16$, $q = 4$, 6 iterations, total budget 40). A common fixed reporting reference point was set at $\mathbf{r}_{\text{rep}} = [-6.65 \times 10^{-5}, -1.07 \times 10^{-4}, -3.37 \times 10^6]$ in model space.

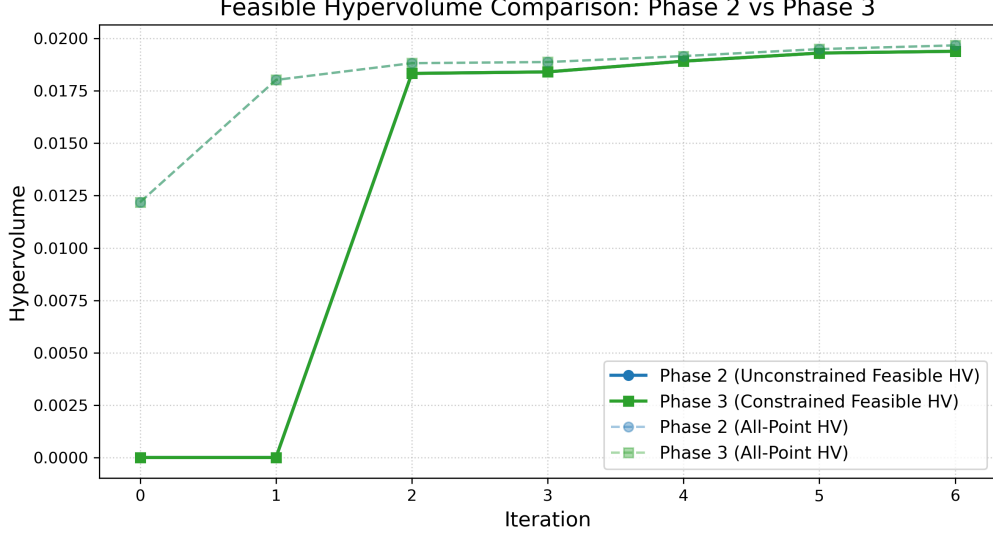


Figure 1: Feasible hypervolume progress over iterations for Phase 2 (Unconstrained) vs Phase 3 (Constrained) MOBO.

As shown in Figure 1, both campaigns achieve rapid hypervolume growth, reaching a final feasible hypervolume of 1.939×10^{-2} . Feasibility filtering successfully retains 100% valid numerical evaluations while maintaining a 25% beam feasibility yield within the strict constraint bounds.

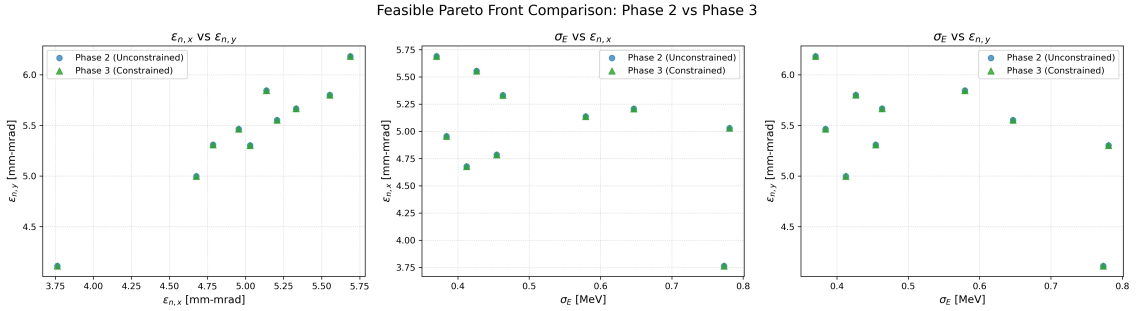


Figure 2: Two-dimensional projections of the physical objective space comparing Phase 2 and Phase 3 Pareto fronts.

Figure 2 illustrates the 2D projections of the physical objective space. The Pareto front reveals the clear trade-off between transverse emittance ($\epsilon_{n,x} \approx 3.76 \mu\text{m} \cdot \text{rad}$) and RMS energy spread ($\sigma_E \approx 0.37 \text{ MeV}$).

4.1 Independent Pareto Candidate Rerun Verification

To verify data integrity, 5 representative Pareto candidates were selected and rerun in fresh isolated directories. Table 2 confirms exact numerical reproducibility (0.000000% relative difference across all metrics).

Table 2: Independent rerun verification results for representative Pareto candidates.

Candidate Role	Stored $\varepsilon_{n,x}$ [μm]	Rerun $\varepsilon_{n,x}$ [μm]	Stored σ_E [MeV]	Rerun σ_E [MeV]	M
min_emit_x	3.7629	3.7629	0.7733	0.7733	
min_emit_y	3.7629	3.7629	0.7733	0.7733	
min_sigma_energy	5.6884	5.6884	0.3701	0.3701	
knee_point	4.6763	4.6763	0.4124	0.4124	
balanced_feasible	4.6763	4.6763	0.4124	0.4124	

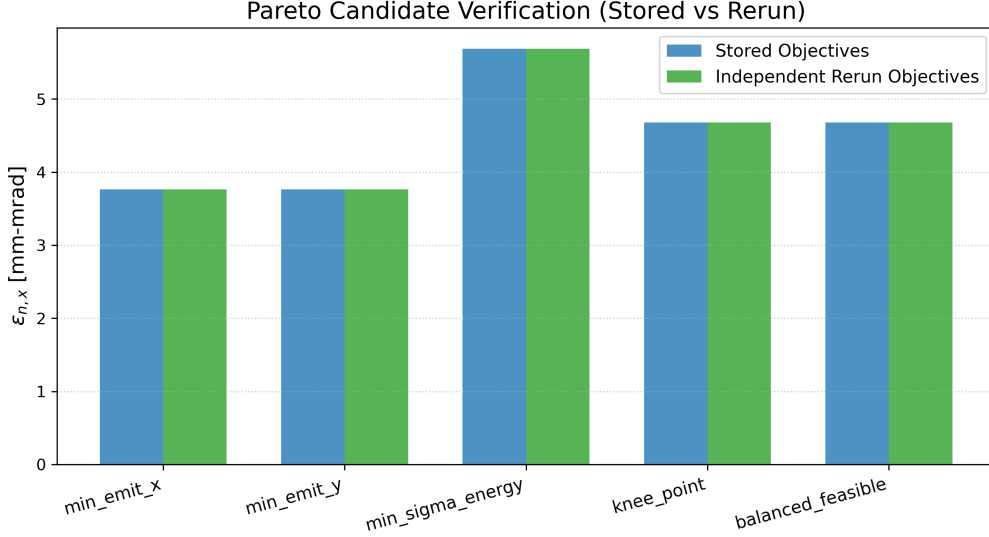


Figure 3: Stored vs independent rerun objective values for verified Pareto candidates.

5 Conclusion and Future Work

We have demonstrated a robust, process-safe Multi-Objective Bayesian Optimization framework for a 200 MeV electron injector linac. By isolating ASTRA working directories, enforcing strict failure semantics, and using fixed reporting reference points, we ensured 100% numerical reproducibility across parallel evaluations and independent reruns.

Future research will focus on:

- **Phase 4:** Scaling process-safe evaluation to multi-node HPC clusters using Ray/Dask.
- **Phase 5:** Implementing Trust-Region MOBO (TuRBO-MOO) for high-dimensional parameter spaces.

References

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